Mohammed Dasouqi

Riyadh, Saudi Arabia | +966 53 862 2526 | mohammeddasouqi@hotmail.com |
linkedin.com/in/mohammeddasouqi | github.com/7amoody1985 | dasouqi.com

Transferable Iqama — available to join without new visa sponsorship

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on, end-to-end experience across the machine learning lifecycle — from data preprocessing and model training to evaluation and deployment. Built deep learning systems for medical image analysis (PyTorch, YOLO, EfficientUNet++), NLP and semantic-search pipelines (Sentence-BERT, LlamaIndex, local LLM deployment), and data-driven backend applications (FastAPI, SQL). Strong foundation in data structures, algorithms, object-oriented software design, and software engineering principles. Seeking an AI/ML Engineer or Data Engineer role building scalable, production-grade intelligent systems.

EDUCATION

University of Nottingham
Bachelor of Science (Honours), Software Engineering
September 2021 – July 2025

- Classification: Upper Second Class Honours (2:1)

TECHNICAL SKILLS

- **Machine Learning & AI:** PyTorch, YOLO (object detection), EfficientUNet++ (image segmentation), Sentence-BERT, NLP question-answering models, LlamaIndex, Ollama (local LLM deployment), semantic embeddings, model evaluation (precision, recall, F1-score, Dice, IoU)
- **Programming Languages:** Python, Java, Kotlin, C++, PHP, JavaScript, SQL, HTML/CSS
- **Backend & Data:** FastAPI, SQLAlchemy, Retrofit, REST API design, asyncio/aiohttp, MySQL, Firebase/Firestore
- **Core Foundations:** Data structures & algorithms, object-oriented design, software development lifecycle (SDLC), design patterns (Observer, Repository, MVVM)
- **Tools & Platforms:** Git/GitHub, TensorBoard, CUDA, Maven, JUnit 5, VS Code, JetBrains IDEs, Google Maps/Places/Directions APIs

PROJECTS

Brain Tumor Detection & Segmentation — Final Year Project | ***PyTorch 2.6, YOLO, RepVGG-GELAN, EfficientUNet++, EfficientNet-B0, CUDA***

- Designed and trained a two-stage deep learning pipeline on the BR35H MRI dataset (800 scans) combining a custom RepVGG-GELAN detector with a nested EfficientUNet++ segmentation network; segmentation model achieved a Dice score of 93.75%, IoU of 89.44%, precision of 95.09%, and recall of 95.92% on the held-out test set
- Detection model met or exceeded the results of the original RepVGG-GELAN paper (Precision 98.6% vs 98.2%, AP50:95 73.0% vs 72.3%), trained entirely from scratch without pretrained detection weights
- Implemented annotation quality filtering — excluding imprecise circular and elliptical ground truth masks from training — which improved segmentation Dice score by 1.5% and eliminated a systematic 'circular effect' bias in predicted masks
- Built the full ML engineering pipeline from scratch: dataset conversion scripts (YOLO format), custom detection-to-segmentation cropping bridge, joint data augmentation, TensorBoard logging, and per-epoch visual validation of predicted vs. ground truth masks

Intelligent Web Crawler (AI Agent) | ***Python, asyncio/aiohttp, BeautifulSoup, Sentence-BERT, NLP***

- Built a budget-aware asynchronous web crawler combining Sentence-BERT semantic scoring, anchor-text relevance weighting, and a BERT-based QA snippet extraction pipeline to dynamically prioritize URLs, replacing static breadth-first traversal strategies

- Implemented a closed-loop online fine-tuning system that periodically re-trained both the embedding and snippet-extraction models on collected data mid-crawl, improving relevance scores in 80% of experimental runs
- Benchmarked across 25 runs, achieving average precision of 95.4%, recall of 85.6%, and F1-score of 90.1%, outperforming breadth-first and TF-IDF baseline crawlers while maintaining semantic diversity above 60% throughout all runs

Chat SQL Agent (AI-Powered Web App) | ***Python, FastAPI, LlamaIndex, Ollama, SQLAlchemy***

- Built a natural-language-to-SQL application that lets non-technical users query relational databases through a RESTful FastAPI backend, removing the need to write SQL manually
- Integrated LlamaIndex with a locally hosted Ollama LLM for schema reasoning and adaptive SQL generation, enabling fully offline, context-aware query translation
- Implemented semantic embeddings for local inference, removing dependency on external API calls and eliminating per-query API costs

AutoConnect (Android App) | ***Kotlin, Jetpack Compose, Firebase Firestore, Retrofit, Google Maps/Directions/Places API***

- Designed and built a feature-complete roadside-assistance Android app with a 4-layer service-oriented architecture (Data, Service, UI, Utility), 8 screens, and role-based access control for customers, drivers, and admins
- Implemented real-time driver-to-customer location tracking via Firestore listeners (Observer pattern) with 30-second background location sync, plus push notifications for live tow-request status updates
- Integrated Retrofit-based REST clients for Google Directions, Places, and Maps SDKs within a Repository pattern (single source of truth), separating all external API interactions from the UI layer
- Built the full UI in Jetpack Compose (no XML) with reusable components (CustomTopAppBar, MapViewComponent, TechnicianGrid) and a ViewModel layer, enabling clean separation of state management from rendering logic

Personal Portfolio Website | ***HTML, CSS, JavaScript (Vanilla — no frameworks), Cloudflare Workers, Anthropic Claude API***

- Built a fully custom single-page portfolio from scratch using only vanilla HTML, CSS, and JavaScript, no frameworks, no build step, no dependencies, with a content-driven architecture where a single configuration object renders the entire site
- Designed and deployed a serverless AI assistant (Anthropic Claude via a Cloudflare Worker proxy) that answers questions about my CV and projects, grounded in a curated knowledge base and project-report digests, with rate limiting, daily cost caps, and Turnstile bot protection
- Developed three interactive in-browser demos with zero backend: an animated MRI segmentation pipeline (detect → crop → segment) visualizing the Brain Tumor Detection FYP, a semantic web-crawler simulation, and a natural-language-to-SQL chat

Brick Breaker Game (Refactor & Extension) | ***Java 17, JavaFX, Maven, JUnit 5***

- Inherited a buggy JavaFX codebase and resolved 40+ bugs including race conditions on simultaneous block destruction, inconsistent ball physics, and broken hitbox detection at high speeds
- Refactored a monolithic structure into a focused package architecture, added JUnit 5 test coverage, thread synchronization, and extended the game with a full UI system including menus, sound, and save/load persistence

EXPERIENCE

Automotive Diagnostics Technician — ***Digital Engineering, Riyadh, Saudi Arabia***

May 2026 (1 month)

- Diagnosed vehicle electronic and software faults using OBD-II diagnostic tools and manufacturer software, identifying root causes across multiple vehicle systems
- Performed ECU parameter calibration and software updates to bring vehicle systems back to manufacturer specification
- Applied a systematic, evidence-based troubleshooting methodology, directly transferable to debugging and root-cause analysis in software and ML systems

LANGUAGES

- Arabic — Native | English — Fluent (bilingual)